# John Parks

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## EDUCATION

**University of Cincinnati, College of Engineering and Applied Science, University Honors Program** Cincinnati, OH
*CEAS Bachelor of Science, Major: Computer Science, CEAS Dean's List, GPA: 3.778*
*Expected Graduation: April 2026*

## WORK EXPERIENCE

**UCdasec Lab – University of Cincinnati,** Cincinnati, OH May 2025 – Present
*Hardware Security Researcher*

- Created a static analysis tool to detect hardware Common Weakness Enumerations in RTL code
- Built parsing and analysis pipelines to extract and analyze structural patterns from hardware RTL code

**VISENS Research Group - University of the Basque Country,** Bilbao, Spain Jan 2025 – May 2025
*Artificial Intelligence Researcher*

- Researched AI-based spasticity detection using BlazePose and computer vision frameworks
- Designed a Python-based anomaly detection algorithm to identify and correct pose estimation errors

**Pole/Zero,** West Chester Township, OH May 2024 – August 2024
*Software Engineer Co-op*

- Developed and executed comprehensive test plans to ensure software reliability and compliance with ISO standards
- Designed and implemented .NET applications to migrate thousands of data files, ensuring scalability and data integrity
- Upgraded shared codebase across multiple software packages, improving system performance, scalability, and security
- Modified existing applications to support multiple versions of spectrum analyzers, saving the company $180,000

**London Computer Systems,** Cincinnati, OH August 2023 – December 2023, January 2023 – April 2023
*Full Stack Developer Co-op*

- Delivered full-stack web development solutions, enhancing application functionality and user experience
- Wrote and optimized SQL scripts to update databases while maintaining accuracy, integrity, and performance
- Collaborated with cross-functional teams in Agile environments, meeting deadlines and requirements
- Presented new features to product managers and stakeholders, ensuring alignment with business goals

**BARK Handmade,** Cincinnati, OH May 2020 – March 2023
*Co-founder*

- Co-founded a woodworking startup, overseeing end-to-end product design, manufacturing, and delivery
- Managed financial operations, including budgeting, expense tracking, and payment planning
- Directed supply chain and material sourcing strategy, ensuring timely delivery of materials and end products

## PROJECTS

**Sudoku Solver** (Python, PyQt5)

- Created a backtracking algorithm that solves Sudoku puzzles by detecting constraint violations
- Built an interactive GUI enabling users to create and solve custom Sudoku puzzles
- Extended the solving algorithm to support advanced constraints such as Kropki dots, killer cages, thermometers, etc.

**AI Healthcare Bot** (AWS, Python, Angular, TypeScript, HTML, CSS)

- Fine-tuned a MedAlpaca LLM for disease classification using a symptom-disease dataset and deployed it on AWS
- Improved disease classification accuracy from 7.8% to 51.4% (6.6x increase) on 3,000 evaluation samples
- Designed and implemented a web-based UI allowing users to interact with the deployed model

## PUBLICATIONS

**Parks, J., et al.** "PickyRTL: A Static Analysis Tool to Detect Hardware CWEs at the Register-Transfer Level." *IEEE SaTC 2026* (Accepted)

## SKILLS

- **Languages:** Python, JavaScript/TypeScript, SQL, VB.NET, C#, C++, HTML/CSS
- **Technologies/Frameworks:** Git, NodeJS, Angular, GraphQL, REST API, BlazePose, AWS, GitHub Actions, Agile Development